

[illegible]

Total marks – 120

Attempt Questions 1 - 8

All questions are of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.	Marks
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(a) Find $\int \cos x \sin^6 x \, dx$	1
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(b) Evaluate $\int_0^1 \frac{2+6x}{\sqrt{4-x^2}} \, dx$, leaving your answer in exact form	3
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(c) Use integration by parts to evaluate $\int_0^{\sqrt{3}} x \tan^{-1} x \, dx$	4
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(d) (i) Find real constants A and B such that $\frac{3}{(x-2)(2x-1)} = \frac{A}{x-2} + \frac{B}{2x-1}$	2
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(ii) Hence find $\int \frac{3 \, dx}{(x-2)(2x-1)}$	2
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(e) Using the substitution $t = \tan \frac{\theta}{2}$ and the results of (d), find $\int \frac{5}{4-3\sin \theta} \, d\theta$	3
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(a) Let $z = \frac{3-6i}{2+i}$, find

(i) $|z|$

2

(ii) $\arg z$

2

(b) Find real values p and q where $\frac{p-5qi}{1+i} = \overline{1-4i}$

3

(c) Let $u = \frac{7\sqrt{2}}{2}(1+i)$, $v = r \cos \theta + ir \sin \theta$ and $uv = 42 \left(\cos \frac{\pi}{20} + i \sin \frac{\pi}{20} \right)$

(i) Write u in modulus-argument form.

2

(ii) Find r and θ .

2

(d) z lies on the locus defined by $|z+2|=2$ and let $\arg z = \theta$

(i) By use of an appropriate diagram, show that $\arg(z+2) = 2\theta - \pi$

2

(ii) Hence, or otherwise, find $\arg(z^2 + 6z + 8)$

2

- (a) Consider the rectangular hyperbola $x^2 - y^2 = 4$.
- (i) Sketch the curve, showing the coordinates of the foci S and S' and the equations of the directrices and asymptotes. 3
- (ii) The point $P(2 \sec \theta, 2 \tan \theta)$ lies on the curve. 2
Show that the tangent at P has equation $x \sec \theta - y \tan \theta = 2$.
- (iii) The tangent meets the $\tilde{x}$ -axis at Q . 3
Show that the locus of the midpoint M of PQ is given by $x^2 - y^2 - 3 = \frac{1}{y^2 + 1}$
- (b) The polynomial $u(x) = mx^7 + nx^6 + 1$ is divisible by $(x+1)^2$.
- (i) Show that $7m = 6n$. 1
- (ii) Find the values of m and n , where m and n are real numbers. 2
- (c) Given that α, β and γ are the roots of the equation $x^3 + 3x + 1 = 0$
- (i) Find a polynomial equation of smallest degree that has α^2, β^2 and γ^2 as roots. 2
- (ii) Hence find $\alpha^2 + \beta^2 + \gamma^2$ 1
- (d) Which one of the following diagrams below could represent the location of the roots of $z^5 + z^2 - z + c = 0$ in the complex plane, where c is a real number. 1
Without any calculations, justify your answer.

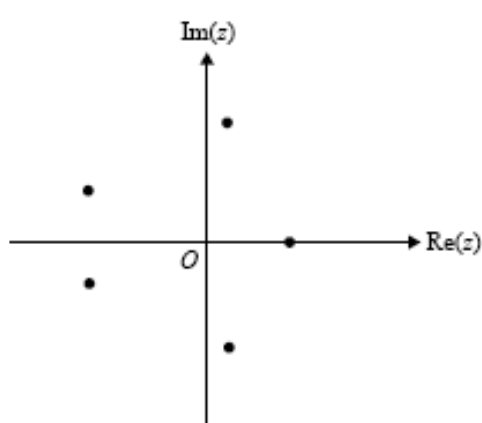


Diagram A

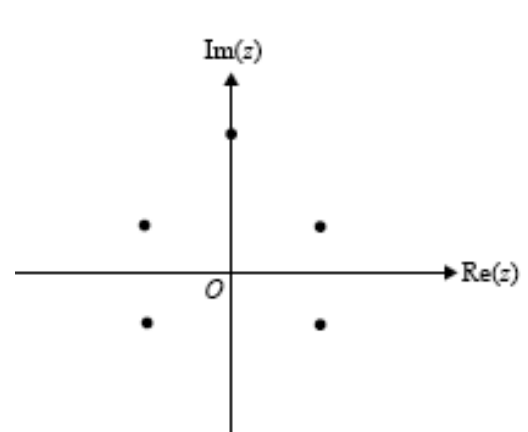
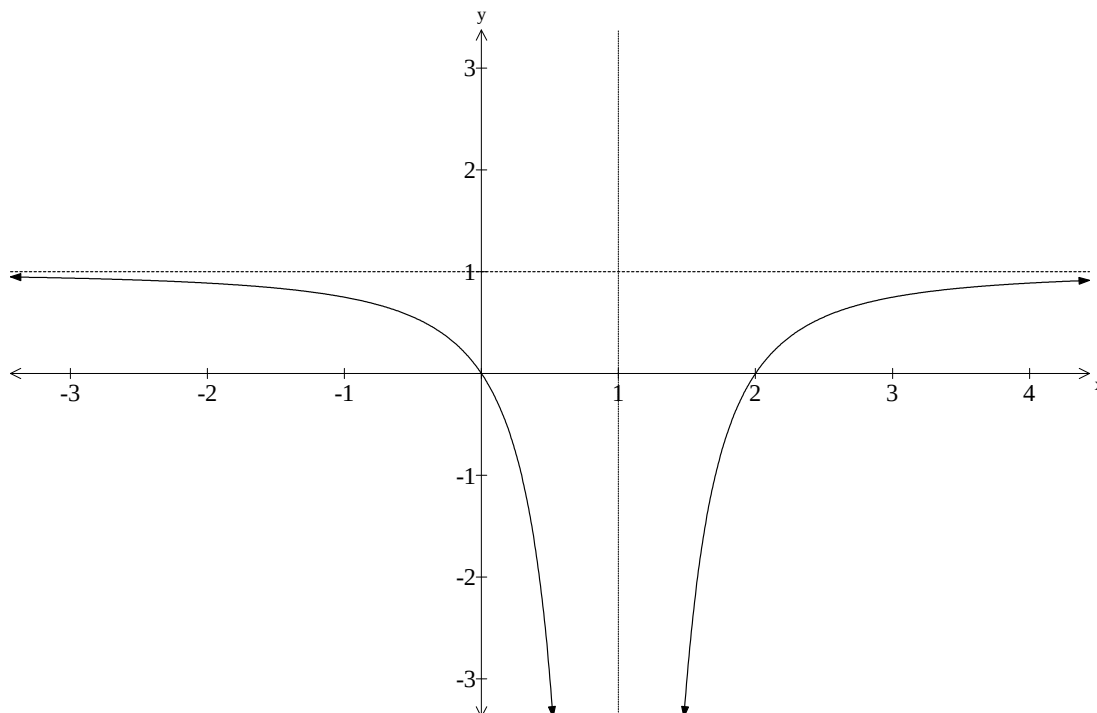


Diagram B

- (a) The graph below shows a function that has $\tilde{x}$ -intercepts at $x = 0$ and $x = 2$. There is a vertical asymptote at $x = 1$ and a horizontal asymptote of $y = 1$. The graph is symmetrical about the line $x = 1$.



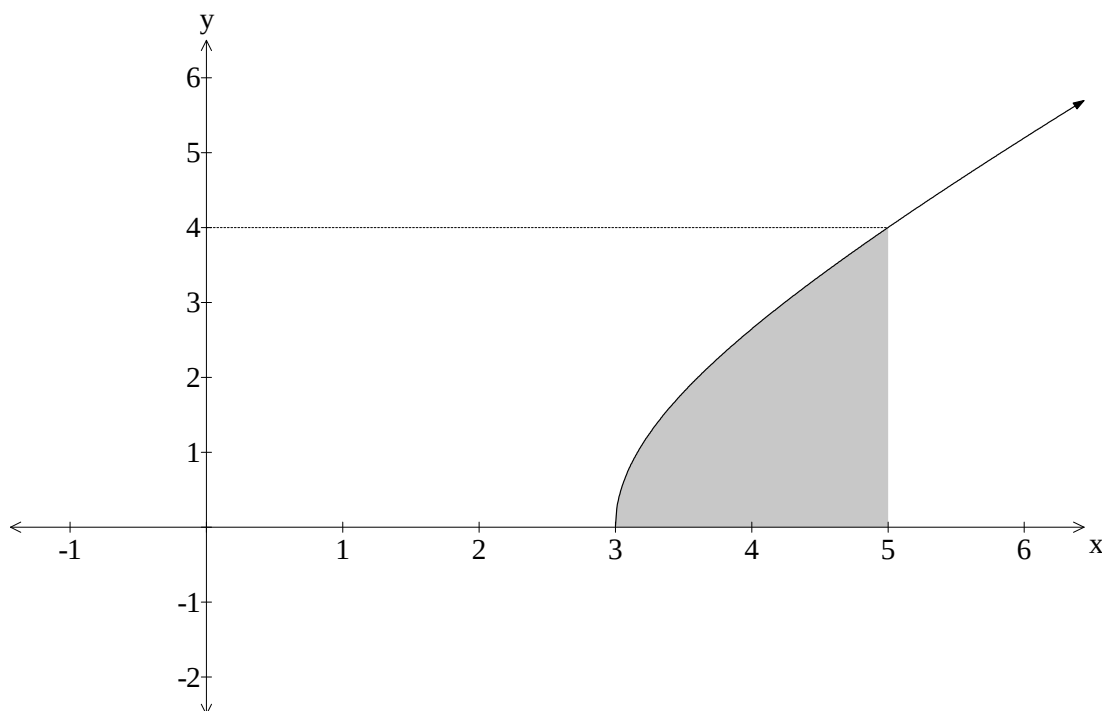
Without using calculus, sketch the following graphs on the ANSWER sheet provided on page 15, clearly showing any asymptotes and intercepts.

- | | | |
|-------|----------------------|----------|
| (i) | $y = f(x - 1)$ | 1 |
| (ii) | $y = [f(x)]^2$ | 2 |
| (iii) | $y^2 = f(x)$ | 2 |
| (iv) | $y = \tan^{-1} f(x)$ | 2 |

Question 4 continues on page 6

(b) The graph of $f(x) = \sqrt{x^2 - 9}$ is shown below.

The area between $f(x)$ and the x -axis for $3 \leq x \leq 5$ is shaded.



(i) Using the method of shells, show that the volume, V , of the solid formed when the shaded area is rotated about the y -axis is given by

2

$$V = \int_3^5 2\pi x (x^2 - 9)^{\frac{1}{2}} dx$$

(ii) Hence calculate the volume.

1

(c) (i) Given $f(x) = f(a - x)$ and using the substitution $u = a - x$, prove that

3

$$\int_0^a x f(x) dx = \frac{a}{2} \int_0^a f(x) dx$$

(ii) Hence, or otherwise, prove that $\int_0^\pi F(x) dx = \frac{\pi^2}{4}$, if $F(x) = \frac{x \sin x}{1 + \cos^2 x}$

2

- (a) (i) Given $I_n = \int_0^1 x^n e^{2x} dx$, where n is a positive integer, show that 2

$$I_n = \frac{1}{2}(e^2 - nI_{n-1})$$

- (ii) Hence evaluate $\int_0^1 x^3 e^{2x} dx$ 3

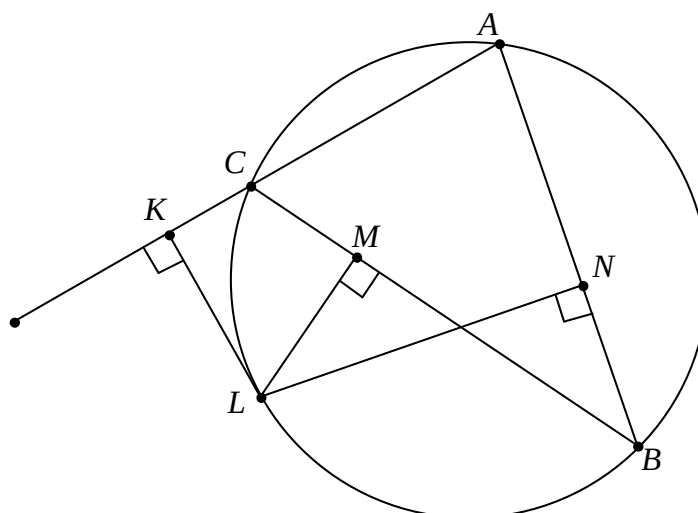
- (b) Fifteen new students at NSGHS are distributed evenly among the classes of Miss V, Mr. S and Ms L.

Given that there are three children with red hair among the fifteen and that the students are distributed randomly, find:

- (i) the number of ways that all the children with red hair end up in the same class. 2
- (ii) the probability that each class gets one child with red hair. 2

- (c) The diagram below shows triangle ABC inscribed in a circle with L a point on the arc BC .

LK is perpendicular to AC produced and LN is perpendicular to AB .



- (i) Copy the diagram into your Answer book
- (ii) Explain why $CKLM$ and $MNBL$ are cyclic quadrilaterals. 2
- (iii) Explain why $\angle KCL = \angle ABL$. 1
- (iv) Hence, or otherwise, prove that K , M and N are collinear. 3

Question 6 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(3x - 2)$

4

(b) Given that $\sin\left(\frac{1}{2}y\right) = \frac{1}{2}(x^2 - 2)$ and that $x > 0$ and $y > 0$.

3

Show by differentiating implicitly that $\frac{dy}{dx} = \frac{4}{\sqrt{4-x^2}}$

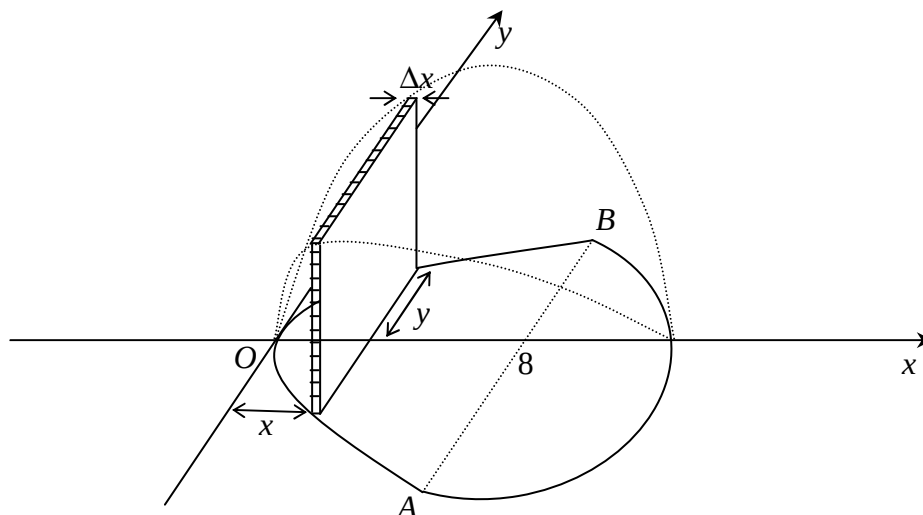
(c) The diagram below shows a solid with its base in the x - y plane.

Every cross-section perpendicular to the x -axis is a square.

One part of the base is the segment OAB of the parabola $y^2 = 2x$ cut off by the line $x = 8$.

The other part of the base is a semi-circle with diameter AB .

Consider a slice S , perpendicular to the x -axis, of width Δx .



(i) Find the coordinates of B and hence find the distance AB .

2

(ii) Show that the volume ΔV of S is given by $\Delta V \approx 8x\Delta x$ for $0 \leq x \leq 8$.

2

(iii) By first finding an expression for ΔV of S when $x > 8$, calculate the volume of the solid.

4

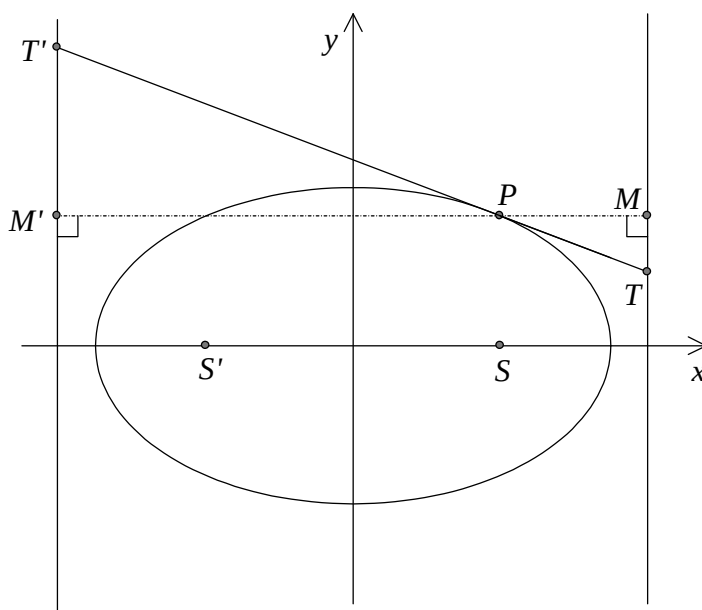
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The diagram below shows an ellipse $b^2x^2 + a^2y^2 = a^2b^2$, where S and S' are the foci.

The diagram shows a tangent at $P(a \cos \theta, b \sin \theta)$, intersecting the two directrices at T and T' .

M and M' are the foot of the perpendiculars drawn from P to their respective directrices.



(a) Show that $SP + S'P = 2a$. 2

(b) You may assume that the tangent at P is $xb \cos \theta + ya \sin \theta = ab$. (Do **NOT** prove this)

Let $\alpha = \angle SPT$ and $\beta = \angle S'PT'$

(i) Show that T has coordinates $\left(\frac{a}{e}, \frac{b(e - \cos \theta)}{ae \sin \theta} \right)$ 1

(ii) Show that $\angle PST = 90^\circ$ 2

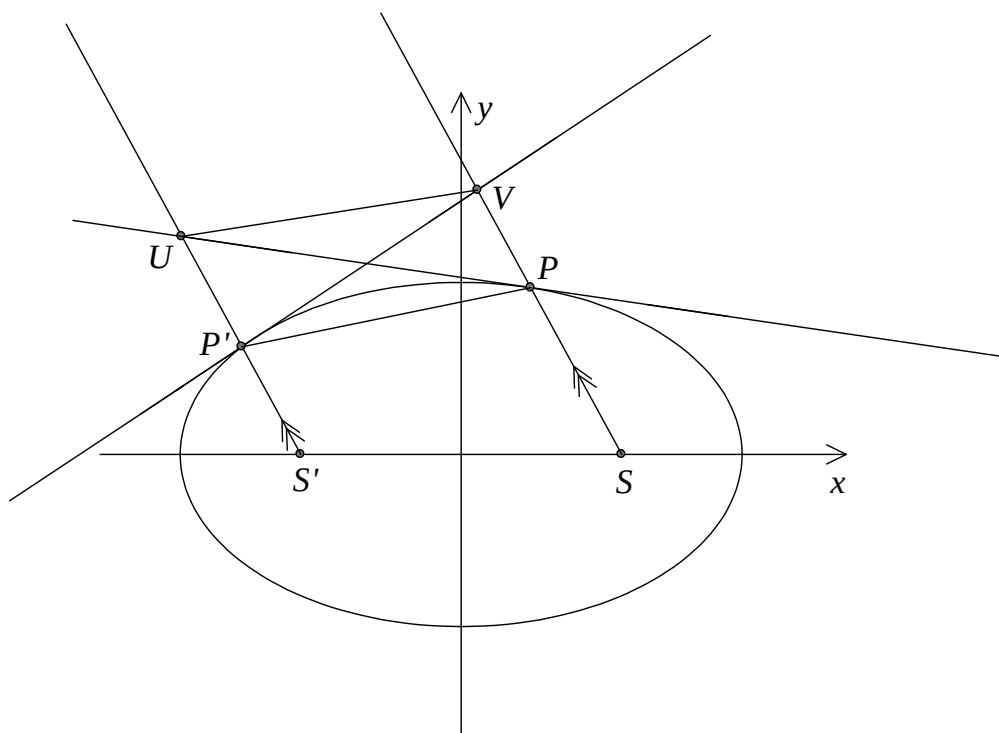
(iii) Show that $\frac{PM}{PT} = \frac{PM'}{PT'}$ 1

(iv) Deduce that $\alpha = \beta$. 2

Question 7 continues on page 11

(c) Consider the diagram below, where $SV \parallel S'U$.

The tangent at P intersects the ray $S'U$ at U and the tangent at P' intersects the ray SV at V .



- | | | |
|-------|---|----------|
| (i) | Copy the diagram into your Answer booklet. | |
| (ii) | Using (b) show that $\triangle UPS'$ is isosceles. | 3 |
| (iii) | Using (ii) above and also (a), show that $VP = UP'$. | 3 |
| (iv) | Deduce that $UV \parallel PP'$ | 1 |

- (a) (i) Show that $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$, where n is a positive integer. **2**

- (ii) Given that $\tan(\tan^{-1}x + \tan^{-1}y) = \frac{x+y}{1-xy}$, where x and y are real numbers, **1**
 explain why when $x > 1, y > 1$ that $\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$.

- (iii) Hence, or otherwise, show that for $n \geq 1$ **3**

$$\sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \frac{3\pi}{4} + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right)$$

- (iv) Hence write down $\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{2}{r^2}\right)$ **1**

- (b) Let $T_n(x) = \frac{{}^nC_0}{x} - \frac{{}^nC_1}{x+1} + \frac{{}^nC_2}{x+2} - \dots + (-1)^n \frac{{}^nC_n}{x+n}$ for a given integer n and all real x

- (i) If $S_k(x) = \frac{k!}{x(x+1)(x+2)\dots(x+k)}$ where k is an integer, show that **2**

$$S_k(x) - S_k(x+1) = S_{k+1}(x)$$

- (ii) Hence prove using mathematical induction that for $n \geq 1$ **4**

$$T_n(x) = \frac{n!}{x(x+1)(x+2)\dots(x+n)}$$

NOTE: you may use without proof the result ${}^{m+1}C_r = {}^mC_r + {}^mC_{r-1}$

- (iii) Hence by a suitable substitution prove that **2**

$$\frac{{}^nC_0}{1} - \frac{{}^nC_1}{3} + \frac{{}^nC_2}{5} - \dots + (-1)^n \frac{{}^nC_n}{2n+1} = \frac{2^n n!}{1 \times 3 \times 5 \times \dots \times (2n+1)}$$

End of paper



**NORTH SYDNEY
GIRLS HIGH SCHOOL**

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Mathematics Extension 2

Sample Solutions

Question 1

$$(a) \quad \int \cos x \sin^6 x \, dx = \frac{\sin^7 x}{7} + C$$

$$\begin{aligned} (b) \quad \int_0^1 \frac{2+6x}{\sqrt{4-x^2}} \, dx &= 2 \int_0^1 \frac{dx}{\sqrt{4-x^2}} - 3 \int_0^1 \frac{-2x \, dx}{\sqrt{4-x^2}} \\ &= 2 \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_0^1 - 3 \times \left[2\sqrt{4-x^2} \right]_0^1 \\ &= 2 \times \frac{\pi}{6} - 6[\sqrt{3}-2] \\ &= 12 + \frac{\pi}{3} - 6\sqrt{3} \end{aligned}$$

$$\begin{aligned} (c) \quad \int_0^{\sqrt{3}} x \tan^{-1} x \, dx &= \int_0^{\sqrt{3}} \frac{d}{dx} \left(\frac{1}{2} x^2 \right) \tan^{-1} x \, dx \\ &= \left[\frac{1}{2} x^2 \tan^{-1} x \right]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \frac{1}{2} x^2 \times \frac{1}{1+x^2} \, dx \\ &= \frac{3}{2} \times \frac{\pi}{3} - \frac{1}{2} \int_0^{\sqrt{3}} \frac{x^2}{1+x^2} \, dx = \frac{\pi}{2} - \frac{1}{2} \int_0^{\sqrt{3}} \frac{(x^2+1)-1}{1+x^2} \, dx \\ &= \frac{\pi}{2} - \frac{1}{2} \int_0^{\sqrt{3}} \left(1 - \frac{1}{1+x^2} \right) \, dx \\ &= \frac{\pi}{2} - \frac{1}{2} \left[x - \tan^{-1} x \right]_0^{\sqrt{3}} = \frac{\pi}{2} - \frac{1}{2} \left(\sqrt{3} - \frac{\pi}{3} \right) \\ &= \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \end{aligned}$$

$$\begin{aligned} (d) \quad (i) \quad \frac{3}{(x-2)(2x-1)} &= \frac{A(2x-1)+B(x-2)}{(x-2)(2x-1)} \\ \therefore A(2x-1)+B(x-2) &= 3 \\ \therefore 2A+B &= 0 \quad [\text{coefficient of } x] \\ \text{Sub } x=2 &\Rightarrow 3A=3 \\ \therefore A=1 &\Rightarrow B=-2 \end{aligned}$$

$$\begin{aligned} (ii) \quad \int \frac{3 \, dx}{(x-2)(2x-1)} &= \int \left(\frac{1}{x-2} + \frac{-2}{2x-1} \right) \, dx \\ &= \int \frac{1}{x-2} \, dx - \int \frac{2}{2x-1} \, dx \\ &= \ln|x-2| - \ln|2x-1| + C \\ &= \ln \left| \frac{x-2}{2x-1} \right| + C \end{aligned}$$

$$(e) \quad t = \tan \frac{\theta}{2} \Rightarrow dx = \frac{2dt}{1+t^2}$$

$$\sin \theta = \frac{2t}{1+t^2}$$

$$\begin{aligned} \int \frac{3}{4-5\sin \theta} d\theta &= \int \frac{3}{4-5\left(\frac{2t}{1+t^2}\right)} \times \frac{2dt}{1+t^2} \\ &= \int \frac{6 dt}{4t^2 - 10t + 4} = \int \frac{3 dt}{2t^2 - 5t + 2} \\ &= \int \frac{3 dt}{(2t-1)(t-2)} \\ &= \ln \left| \frac{t-2}{2t-1} \right| + C \\ &= \ln \left| \frac{\tan^{-1} \frac{\theta}{2} - 2}{2 \tan^{-1} \frac{\theta}{2} - 1} \right| + C \end{aligned}$$

Question 2

(a) $z = \frac{3-6i}{2+i}$

(i) $|z| = \frac{|3-6i|}{|2+i|} = \frac{|3-6i|}{\sqrt{5}} = \frac{3\sqrt{5}}{\sqrt{5}} = 3$

(ii) $z = \frac{3-6i}{2+i} = 3 \times \frac{1-2i}{2+i}$
 $\frac{1-2i}{2+i} = \frac{1-2i}{2+i} \times \frac{2-i}{2-i} = \frac{-5i}{5} = -i$
 $\arg z = \arg(-i) = -\frac{\pi}{2}$

(b) $\frac{p-5qi}{1+i} = \overline{1-4i} \Rightarrow p-5qi = (1+4i)(1+i)$

$\therefore p-5qi = -3+5i$

$\therefore p = -3, -5q = 5$ (Equating real and imaginary parts)

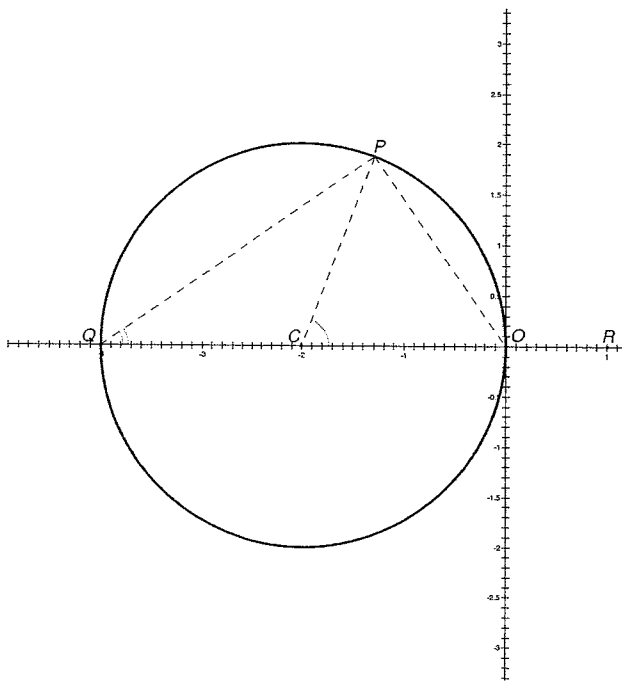
$\therefore p = -3, q = -1$

(c) (i) $u = \frac{7\sqrt{2}}{2}(1+i) = \frac{7\sqrt{2}}{2} \times \sqrt{2} \operatorname{cis} \frac{\pi}{4} = 7 \operatorname{cis} \frac{\pi}{4}$

(ii) $v = \frac{uv}{u} = \frac{42 \operatorname{cis} \frac{\pi}{20}}{7 \operatorname{cis} \frac{\pi}{4}} = 6 \operatorname{cis} \left(\frac{\pi}{20} - \frac{\pi}{4} \right) = 6 \operatorname{cis} \left(-\frac{\pi}{5} \right)$

$\therefore r = 6, \theta = -\frac{\pi}{5}$

- (d) (i) Let z be represented by the point P . Let Q represent the number -4 and C the centre of the circle -2 .



Let $\theta = \arg z \Rightarrow \angle POR = \theta$

$\arg(z+2) = \angle PCO = \pi - 2 \times \angle POC$
 $= \pi - 2 \times (\pi - \theta)$
 $= 2\theta - \pi$

(ii) $\arg(z^2 + 6z + 8) = \arg[(z+2)(z+4)]$
 $= \arg(z+2) + \arg(z+4)$

Now $\arg(z+4) = \angle PQC = \frac{1}{2} \angle PCO$
 (angles at centre and circumference)
 $\therefore \arg(z+4) = \frac{1}{2}(2\theta - \pi) = \theta - \frac{\pi}{2}$
 $\therefore \arg(z^2 + 6z + 8) = 2\theta - \pi + \theta - \frac{\pi}{2}$
 $= 3\theta - \frac{3\pi}{2}$

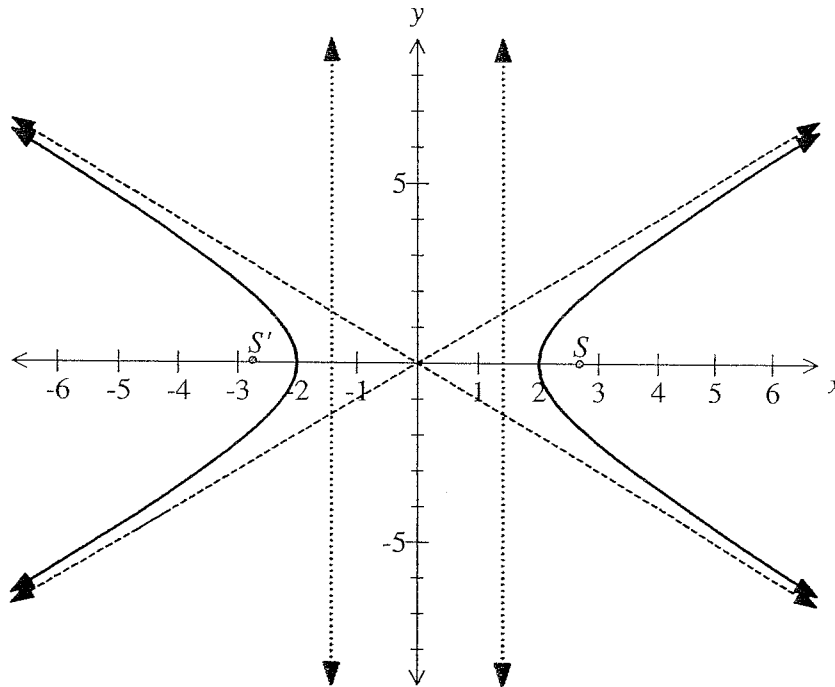
Question 3

(a) (i) $x^2 - y^2 = 4$; $e = \sqrt{2}$

Asymptotes are $y = \pm x$

The directrices are $x = \pm \frac{a}{e} = \pm \frac{2}{\sqrt{2}} = \pm \sqrt{2}$

The foci are at $(\pm ae, 0) = (\pm 2\sqrt{2}, 0)$ i.e. $S(2\sqrt{2}, 0)$ and $S'(-2\sqrt{2}, 0)$



(ii) $x^2 - y^2 = 4 \Rightarrow 2x - 2yy' = 0$

$$\therefore y' = \frac{x}{y} \Rightarrow m = \frac{2 \sec \theta}{2 \tan \theta} = \frac{\sec \theta}{\tan \theta}$$

$$\therefore y - 2 \tan \theta = \frac{\sec \theta}{\tan \theta} (x - 2 \sec \theta) \Rightarrow y \tan \theta - 2 \tan^2 \theta = x \sec \theta - 2 \sec^2 \theta$$

$$\therefore x \sec \theta - y \tan \theta = 2 (\sec^2 \theta - \tan^2 \theta)$$

$$\therefore x \sec \theta - y \tan \theta = 2$$

(iii) $Q: y = 0 \Rightarrow x \sec \theta = 2$

$$\therefore Q(2 \cos \theta, 0) \Rightarrow M(\cos \theta + \sec \theta, \tan \theta)$$

$$\text{LHS} = x^2 - y^2 - 3$$

$$= (\cos \theta + \sec \theta)^2 - (\tan \theta)^2 - 3$$

$$= \cos^2 \theta + 2 + \sec^2 \theta - \tan^2 \theta - 3$$

$$= \cos^2 \theta + 2 + 1 - 3$$

$$= \cos^2 \theta$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\text{RHS} = \frac{1}{y^2 + 1}$$

$$= \frac{1}{(\tan \theta)^2 + 1} = \frac{1}{\tan^2 \theta + 1}$$

$$= \frac{1}{\sec^2 \theta} = \cos^2 \theta$$

So the locus of M is $x^2 - y^2 - 3 = \frac{1}{y^2 + 1}$

Question 3 continued

(b) (i) $u(-1) = u'(-1) = 0$ (Multiple Root Theorem)

$$u'(x) = 7mx^6 + 6nx^5 \Rightarrow u'(-1) = 7m(-1)^6 + 6n(-1)^5 = 0$$

$$\therefore 7m - 6n = 0$$

$$\therefore 7m = 6n$$

(ii) $u(-1) = 0 \Rightarrow m(-1)^7 + n(-1)^6 + 1 = 0$

$$\therefore -m + n + 1 = 0 \Rightarrow n - m = -1 \quad \text{---} (*)$$

From (i) $7m = 6n$

(*) becomes $7n - 7m = -7$ and so $7n - 6n = -7 \Rightarrow n = -7$

$\therefore m = -6$ by substituting into (*) or the result in (*)

$$m = -6, n = -7$$

(c) (i) $x^3 + 3x + 1 = 0$

Let $y = x^2$

$$x^3 + 3x + 1 = 0 \Rightarrow x(x^2 + 3) = -1$$

$$\therefore x^2(x^2 + 3)^2 = 1 \Rightarrow y(y + 3)^2 = 1$$

$$\therefore y^3 + 6y^2 + 9y - 1 = 0$$

(ii) $\alpha^2, \beta^2, \gamma^2$ are the roots of $y^3 + 6y^2 + 9y - 1 = 0$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = -6 \quad (\text{sum of roots})$$

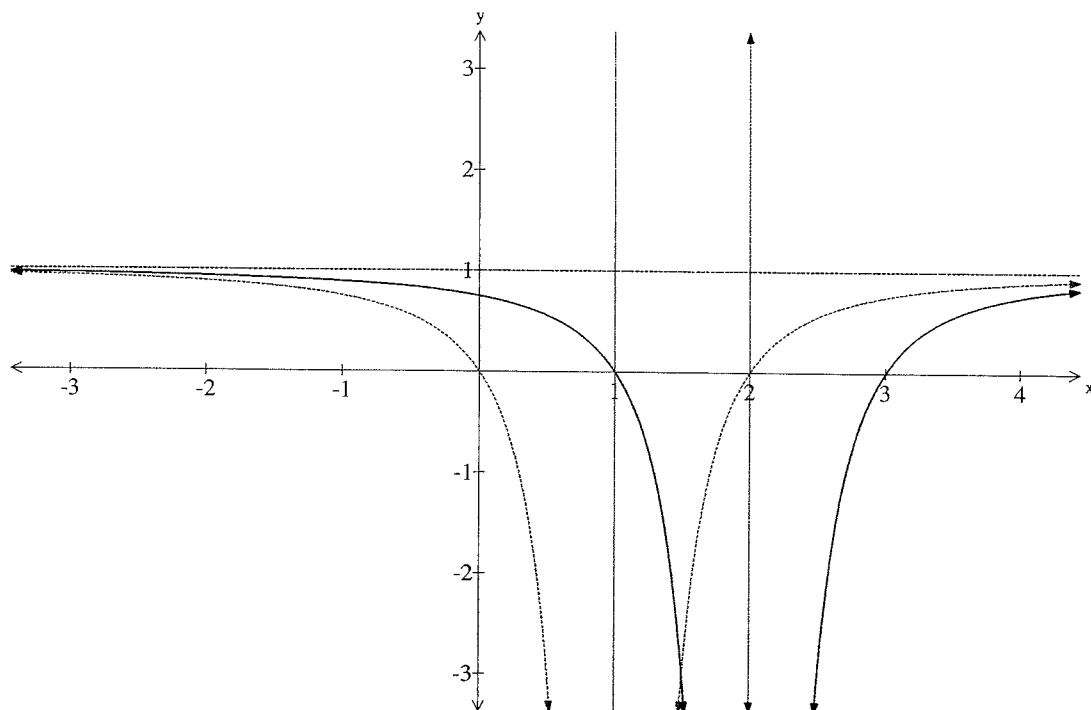
(d) $z^5 + z^2 - z + c = 0$ has real coefficients and so all the roots occur in conjugate pairs.
Diagram B has a root that doesn't have its conjugate pair showing

Answer: Diagram A

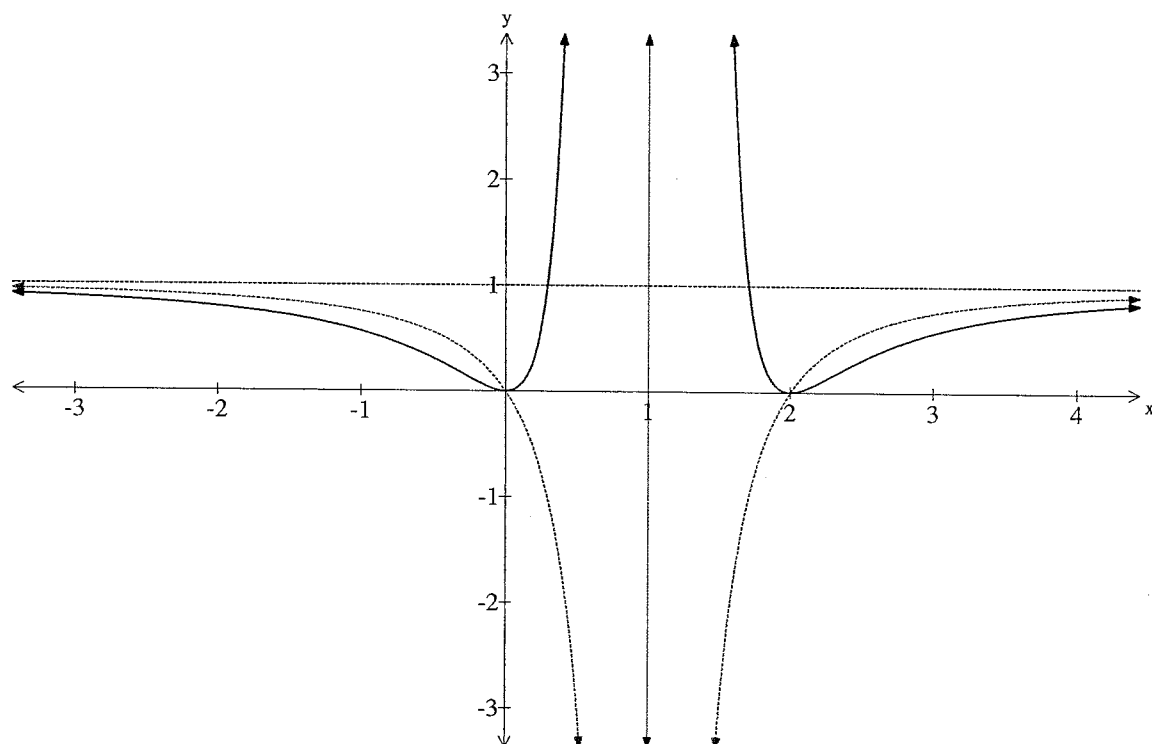
Question 4

(a) ~~Dotted~~ curve existing curve; black curve new transformation

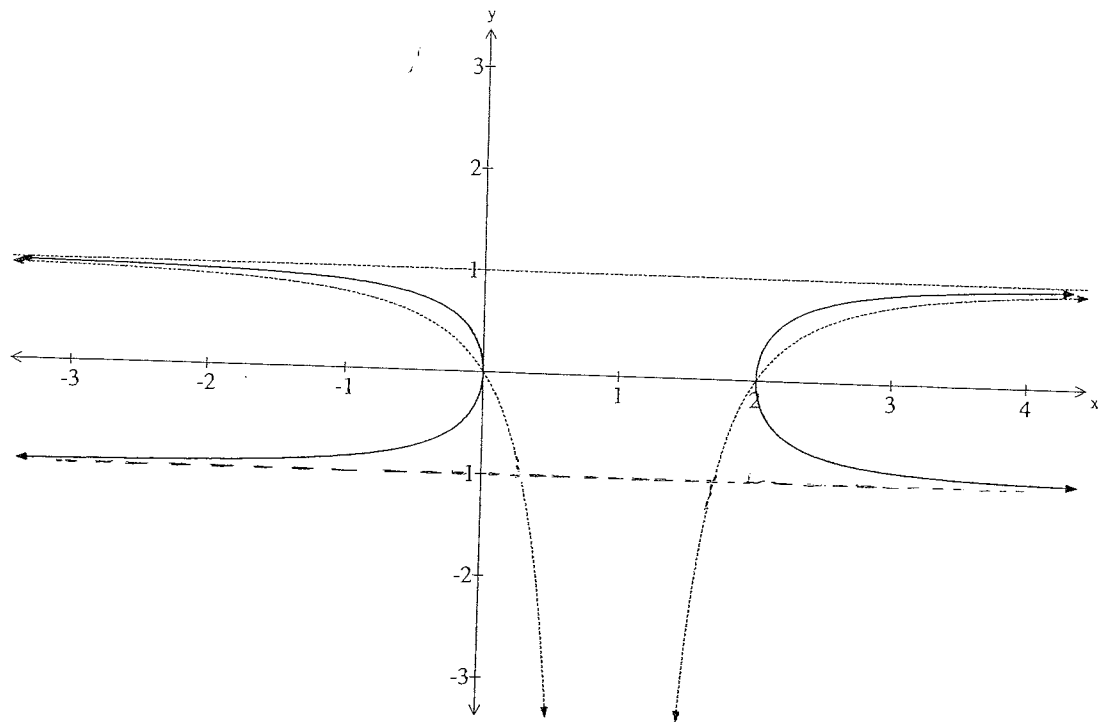
(i) $y = f(x-1)$



(ii) $y = [f(x)]^2$



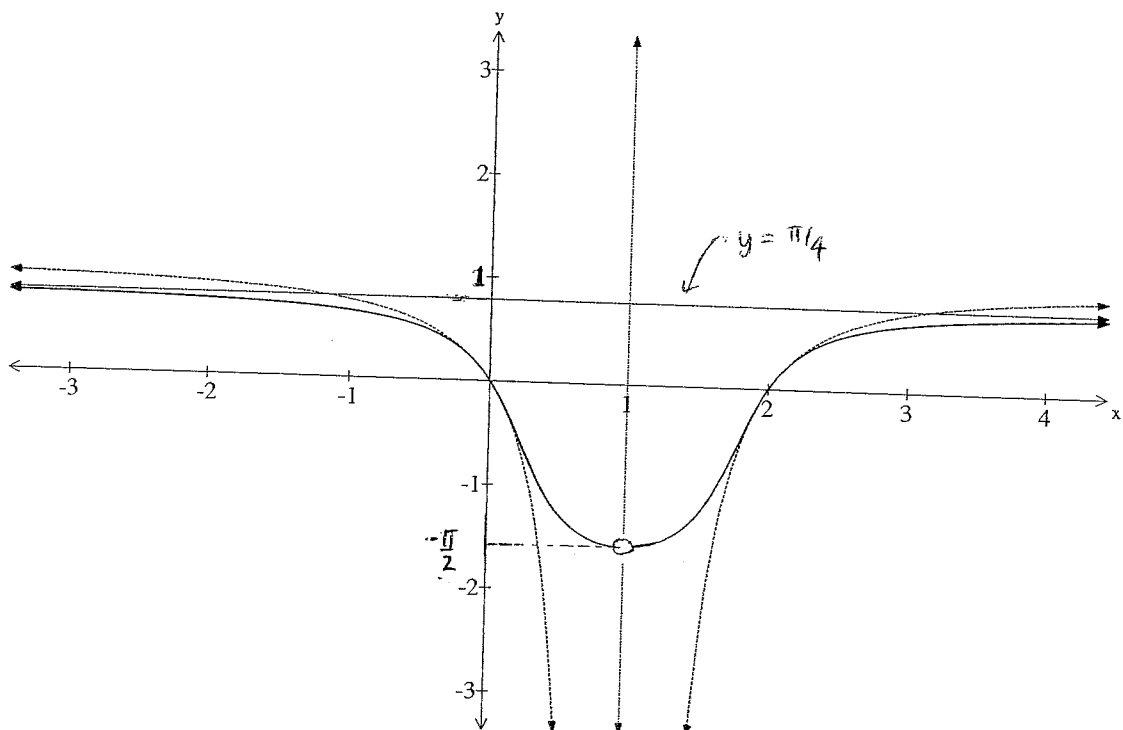
(iii) $y^2 = f(x)$



(iv) $y = \tan^{-1} f(x)$

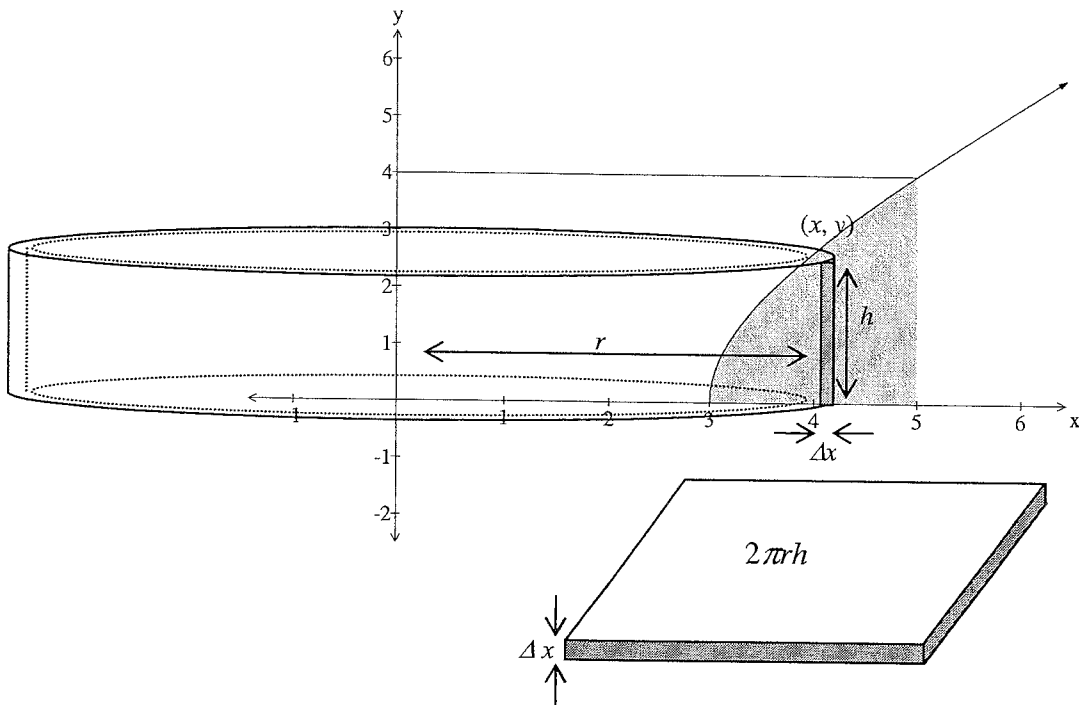
The new horizontal asymptote is $y = \frac{\pi}{4}$.

The curve is not defined at $x = 1$, but it isn't a vertical asymptote



Question 4 continued

- (b) Cut the shell into what is approximately a rectangular prism of length $2\pi r$ and height h .



$$(i) \quad r = x, h = y \Rightarrow \Delta V \approx 2\pi xy$$

$$\therefore \Delta V \approx 2\pi x(x^2 - 9)^{\frac{1}{2}}$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sum_{x=3}^5 \Delta V = \int_3^5 2\pi x(x^2 - 9)^{\frac{1}{2}} dx$$

$$(ii) \quad V\pi = \int_3^5 2x(x^2 - 9)^{\frac{1}{2}} dx$$

$$= \pi \left[\frac{2}{3} (x^2 - 9)^{\frac{3}{2}} \right]_3^5$$

$$= \frac{2\pi}{3} \left(16^{\frac{3}{2}} - 0 \right) = \frac{128\pi}{3} \text{ c.u.}$$

Question 4 continued

(c) (i) $u = a - x \Rightarrow du = -dx$

$$x = a - u$$

$$x = 0 \Rightarrow u = a; x = a \Rightarrow u = 0$$

$$\begin{aligned} \int_0^a x f(x) dx &= \int_a^0 (a-u) f(a-u) (-dx) \\ &= \int_0^a (a-u) f(a-u) du \\ &= \int_0^a (a-u) f(u) du \\ &= \int_0^a a f(u) du - \int_0^a u f(u) du \\ &= \int_0^a a f(x) dx - \int_0^a x f(x) dx \end{aligned}$$

$$\therefore 2 \int_0^a x f(x) dx = \int_0^a a f(x) dx$$

$$\begin{aligned} \int_0^a x f(x) dx &= \frac{1}{2} \int_0^a a f(x) dx \\ &= \frac{a}{2} \int_0^a f(x) dx \end{aligned}$$

(ii)
$$\begin{aligned} \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi}{2} \int \frac{\sin x}{1 + \cos^2 x} dx \\ &= -\frac{\pi}{2} \int \frac{-\sin x}{1 + \cos^2 x} dx \\ &= -\frac{\pi}{2} \left[\tan^{-1}(\cos x) \right]_0^\pi \\ &= -\frac{\pi}{2} \left[\tan^{-1}(-1) - \tan^{-1}(-1) \right] \\ &= -\frac{\pi}{2} \left[-\frac{\pi}{4} - \frac{\pi}{4} \right] \\ &= \frac{\pi^2}{4} \end{aligned}$$

Question 5

$$\begin{aligned}
 \text{(a)} \quad \text{(i)} \quad I_n &= \int_0^1 x^n e^{2x} dx \\
 &= \int_0^1 x^n \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right) dx \\
 &= \left[\frac{1}{2} e^{2x} x^n \right]_0^1 - \int_0^1 \frac{e^{2x}}{2} (n x^{n-1}) dx \\
 &= \frac{1}{2} e^2 - \frac{n}{2} \int_0^1 x^{n-1} e^{2x} dx \\
 &= \frac{1}{2} e^2 - \frac{n}{2} I_{n-1} = \frac{1}{2} (e^2 - n I_{n-1})
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int_0^1 x^3 e^{2x} dx &= I_3 \\
 &= \frac{1}{2} (e^2 - 3 I_2) = \frac{1}{2} e^2 - \frac{3}{2} I_2 \\
 &= \frac{1}{2} e^2 - \frac{3}{2} \left[\frac{1}{2} (e^2 - 2 I_1) \right] \\
 &= \frac{1}{2} e^2 - \frac{3}{4} e^2 + \frac{3}{2} I_1 \\
 &= \frac{1}{2} e^2 - \frac{3}{4} e^2 + \frac{3}{2} \left[\frac{1}{2} (e^2 - I_0) \right] \\
 &= \frac{1}{2} e^2 - \frac{3}{4} e^2 + \frac{3}{4} e^2 - \frac{3}{4} I_0 \\
 &= \frac{1}{2} e^2 - \frac{3}{4} I_0 \\
 &= \frac{1}{2} e^2 - \frac{3}{4} \times \frac{1}{2} (e^2 - 1) \\
 &= \frac{1}{8} e^2 + \frac{3}{8}
 \end{aligned}$$

$$\begin{aligned}
 I_0 &= \int_0^1 e^{2x} dx \\
 &= \frac{1}{2} [e^{2x}]_0^1 \\
 &= \frac{1}{2} (e^2 - 1)
 \end{aligned}$$

Question 5 continued

- (b) (i) Since the classes are “distinguishable” then there are $\binom{3}{1}$ ways of picking the class that has all the red heads. Then the remaining 2 students for that class need to be picked from the remaining 12 students in $\binom{12}{2}$ ways. Then $\binom{10}{5}$ ways to place 5 of the remaining girls in one of the other class, this leaves the last 5 students to be allocated to the remaining class.
i.e. $\binom{3}{1} \times \binom{12}{2} \times \binom{10}{5} = 49\,896$ ways.

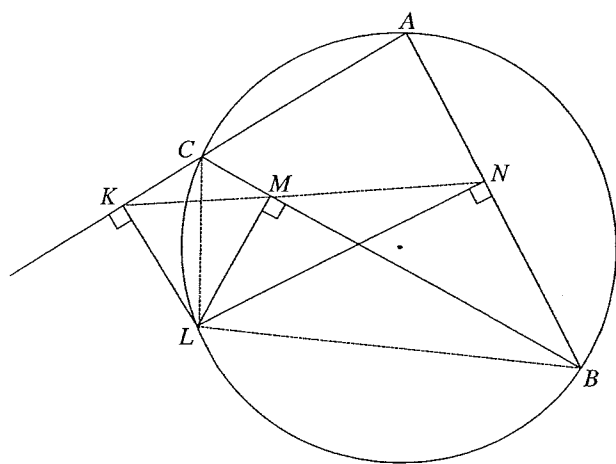
- (ii) Miss V can be allocated one of the redheads in $\binom{3}{1}$ ways and her remaining students in $\binom{12}{4}$ ways. Mr S can be allocated his redhead in $\binom{2}{1}$ ways and the remaining students in $\binom{8}{4}$ ways. The remaining students all go to Ms L’s class. i.e.
 $\binom{3}{1} \times \binom{12}{4} \times \binom{2}{1} \times \binom{8}{4} = 207\,900$ ways.

Without restriction all the students can be allocated in $\binom{15}{5} \times \binom{10}{5}$ ways i.e. in 756 756 ways. So the probability of this happening is $\frac{207\,900}{756\,756} = \frac{25}{91}$.

- (c) (i) Construct KM , MN , LB and CL .

- (ii) In $CKLM$, $\angle CKL = \angle CML = 90^\circ$.

Opposite angles are supplementary and so the quadrilateral is cyclic.



- In $MNBL$, $\angle LMB = \angle LNB = 90^\circ$.

By the converse of angles in the same segment, the quadrilateral is cyclic.

- (iii) $\angle KCL$ is the exterior angle of quadrilateral $ACLB$ and so $\angle KCL = \angle ABL$ by the exterior angle theorem for cyclic quadrilaterals.

- (iv) As $CKLM$ is a cyclic quadrilateral, $\angle KCL = \angle KML$ (angles in same segment)
As $MNBL$ is a cyclic quadrilateral, $\angle LMN = 180^\circ - \angle ABL$ (opposite angles supp.)
From (iii) $\angle KCL = \angle ABL$ and so $\angle LMN = 180^\circ - \angle KML$
 $\therefore \angle KMN = \angle KML + \angle LMN = 180^\circ$
So K , M , and N are collinear.

Question 6

$$\begin{aligned}
 \text{(a)} \quad & \sin(\sin^{-1} x - \cos^{-1} x) = \sin[\sin^{-1}(3x-2)] \\
 & \therefore \sin\left(2\sin^{-1} x - \frac{\pi}{2}\right) = 3x-2 \qquad \left[\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}\right] \\
 & \therefore -\sin\left(\frac{\pi}{2} - 2\sin^{-1} x\right) = 3x-2 \qquad [\sin(-x) = -\sin x] \\
 & \therefore -\cos(2\sin^{-1} x) = 3x-2 \qquad \left[\sin\left(\frac{\pi}{2} - x\right) = \cos x\right] \\
 & \therefore -[1 - 2\sin^2(\sin^{-1} x)] = 3x-2 \qquad \left[\begin{array}{l} \text{Let } \alpha = \sin^{-1} x; \\ \cos 2\alpha = 1 - 2\sin^2 \alpha \end{array}\right] \\
 & \therefore 2x^2 - 1 = 3x-2 \qquad \left[\sin^2(\sin^{-1} x) = [\sin(\sin^{-1} x)]^2\right] \\
 & \therefore 2x^2 - 3x + 1 = 0 \\
 & \therefore (2x-1)(x-1) = 0 \\
 & \therefore x = \frac{1}{2}, 1
 \end{aligned}$$

Now test the solutions in the original equation i.e. $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(3x-2)$

$$x = \frac{1}{2}: \text{LHS} = \sin^{-1} \frac{1}{2} - \cos^{-1} \frac{1}{2} = \frac{\pi}{6} - \frac{\pi}{3} = -\frac{\pi}{6}$$

$$\text{RHS} = \sin^{-1}(3 \times \frac{1}{2} - 2) = \sin^{-1}(-\frac{1}{2}) = -\frac{\pi}{6}$$

$$x = 1: \text{LHS} = \sin^{-1} 1 - \cos^{-1} 1 = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

$$\text{RHS} = \sin^{-1}(3 \times 1 - 2) = \sin^{-1}(1) = \frac{\pi}{2}$$

$$\therefore x = \frac{1}{2}, 1$$

ALTERNATIVELY

$$\text{(b)} \quad \sin\left(\frac{1}{2}y\right) = \frac{1}{2}(x^2 - 2)$$

$$\therefore \frac{1}{2}y' \cos\left(\frac{1}{2}y\right) = x$$

$$y' = \frac{2x}{\cos\left(\frac{1}{2}y\right)} = \frac{2x}{\sqrt{1 - \sin^2\left(\frac{1}{2}y\right)}}$$

$$= \frac{2x}{\sqrt{1 - \left[\frac{1}{2}(x^2 - 2)\right]^2}} = \frac{4x}{\sqrt{4 - (x^2 - 2)^2}}$$

$$= \frac{4x}{\sqrt{4 - (x^2 - 2)^2}}$$

$$= \frac{4x}{\sqrt{x^2(4 - x^2)}} = \frac{4x}{x\sqrt{4 - x^2}} = \frac{4}{\sqrt{4 - x^2}}$$

$$\frac{1}{2}y = \sin^{-1}\left(\frac{x^2}{2} - 1\right)$$

$$\therefore \frac{d}{dx}\left(\frac{1}{2}y\right) = \frac{d}{dx}\sin^{-1}\left(\frac{x^2}{2} - 1\right)$$

$$\therefore \frac{1}{2}y' = \frac{1}{\sqrt{1 - \left(\frac{x^2}{2} - 1\right)^2}} \times x = \frac{x}{\sqrt{1 - \left(\frac{x^4}{4} - x^2 + 1\right)}}$$

$$\therefore y' = \frac{4x}{\sqrt{4x^2 - x^4}} = \frac{4x}{x\sqrt{4 - x^2}} = \frac{4}{\sqrt{4 - x^2}}$$

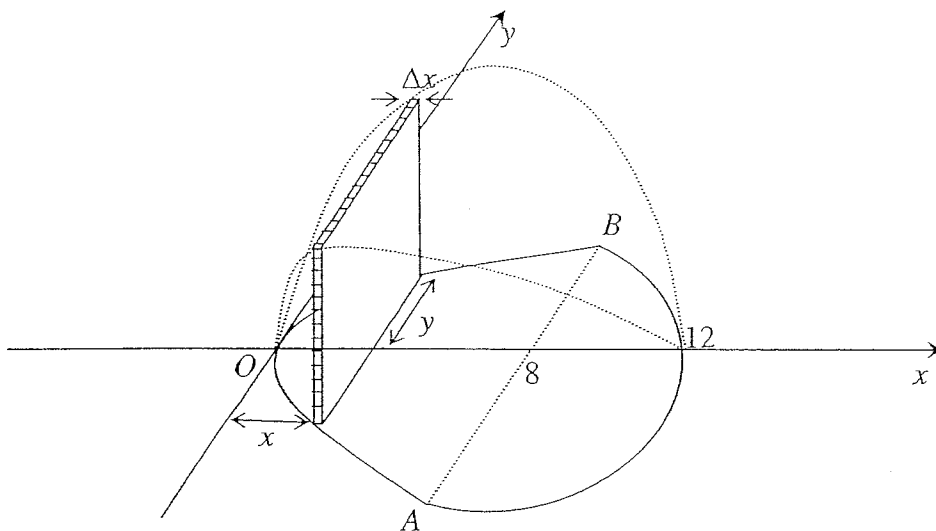
Question 6 continued

(c) (i) $x = 8 \Rightarrow y^2 = 2 \times 8 \Rightarrow y = \pm 4$

$\therefore B(8, 4)$

$\therefore AB = 2 \times 4 = 8$

So the semi-circle has radius 4 and so the extreme x -value is $x = 12$



(ii) The square has side length $2y \Rightarrow \Delta V \approx (2y)^2 \Delta x$

$\therefore \Delta V \approx 4y^2 \Delta x = 4(2x) \Delta x = 8x \Delta x$

(iii) For $8 < x < 12$, the base is $(x-8)^2 + y^2 = 16$

$\therefore \Delta V \approx 4y^2 \Delta x = 4[16 - (x-8)^2] \Delta x = [64 - 4(x-8)^2] \Delta x$

$V = \int_0^8 8x \, dx + \int_8^{12} [64 - 4(x-8)^2] \, dx$

$= \left[4x^2 \right]_0^8 + \left[64x - \frac{4}{3}(x-8)^3 \right]_8^{12}$

$= 256 + \left[\left(768 - \frac{4}{3} \times 64 \right) - (512 - 0) \right]$

$= 426 \frac{2}{3}$

ALTERNATIVELY

With the semi-circular section

$V_{\text{semi-circular}} = 4 \int_8^{12} [16 - (x-8)^2] \, dx$

$= 4 \int_0^4 [16 - x^2] \, dx$

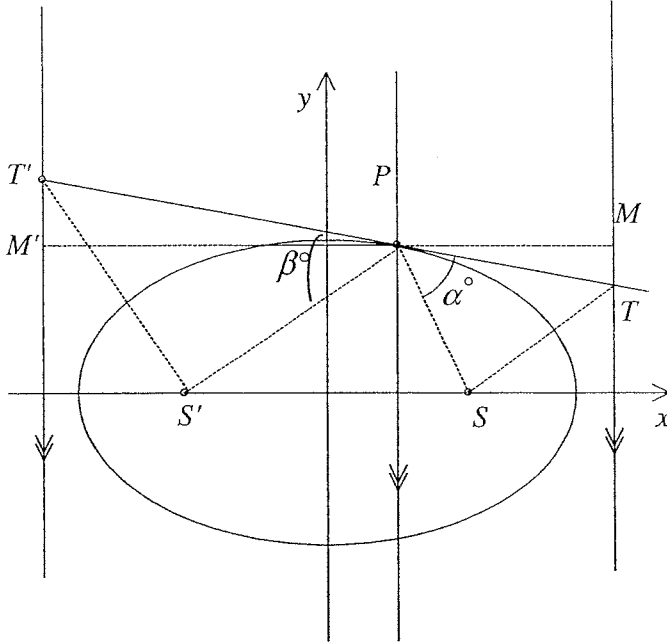
$= 4 \left[16x - \frac{x^3}{3} \right]_0^4$

$= 4 \left(64 - \frac{64}{3} \right)$

$= 4 \left(\frac{2}{3} \times 64 \right) = 170 \frac{2}{3}$

Question 7

- (a) For a conic $SP = ePM$
 $SP + S'P = ePM + ePM'$
 $= e(PM + PM')$
 $= e\left(2 \times \frac{a}{e}\right)$
 $= 2a$



- (b) (i) $T: x = \frac{a}{e} \Rightarrow \left(\frac{a}{e}\right)b \cos \theta + ya \sin \theta = ab$

$$\therefore \frac{b \cos \theta}{e} + y \sin \theta = b$$

$$\therefore y \sin \theta = b - \frac{b \cos \theta}{e} = \frac{b(e - \cos \theta)}{e}$$

$$\therefore y = \frac{b(e - \cos \theta)}{e \sin \theta} \Rightarrow T\left(\frac{a}{e}, \frac{b(e - \cos \theta)}{e \sin \theta}\right)$$

- (ii) $m_{SP} = \frac{b \sin \theta - 0}{a \cos \theta - ae} = \frac{b \sin \theta}{a(\cos \theta - e)}$

$$m_{ST} = \frac{\frac{b(e - \cos \theta)}{e \sin \theta} - 0}{\frac{a}{e} - ae} \times \frac{e \sin \theta}{e \sin \theta} = \frac{b(e - \cos \theta)}{a \sin \theta (1 - e^2)}$$

$$= \frac{b(e - \cos \theta)}{a \sin \theta \times \frac{b^2}{a^2}} \quad \left[e^2 = 1 - \frac{b^2}{a^2} \right]$$

$$= \frac{a(e - \cos \theta)}{b \sin \theta} = -\frac{a(\cos \theta - e)}{b \sin \theta}$$

$$\therefore m_{ST} \times m_{SP} = -1 \Rightarrow \angle PST = 90^\circ$$

Similarly $\angle PS'T' = 90^\circ$

Question 7 continued

(iii) $PM : PM' = PT : PT'$ (parallel lines preserve ratio)

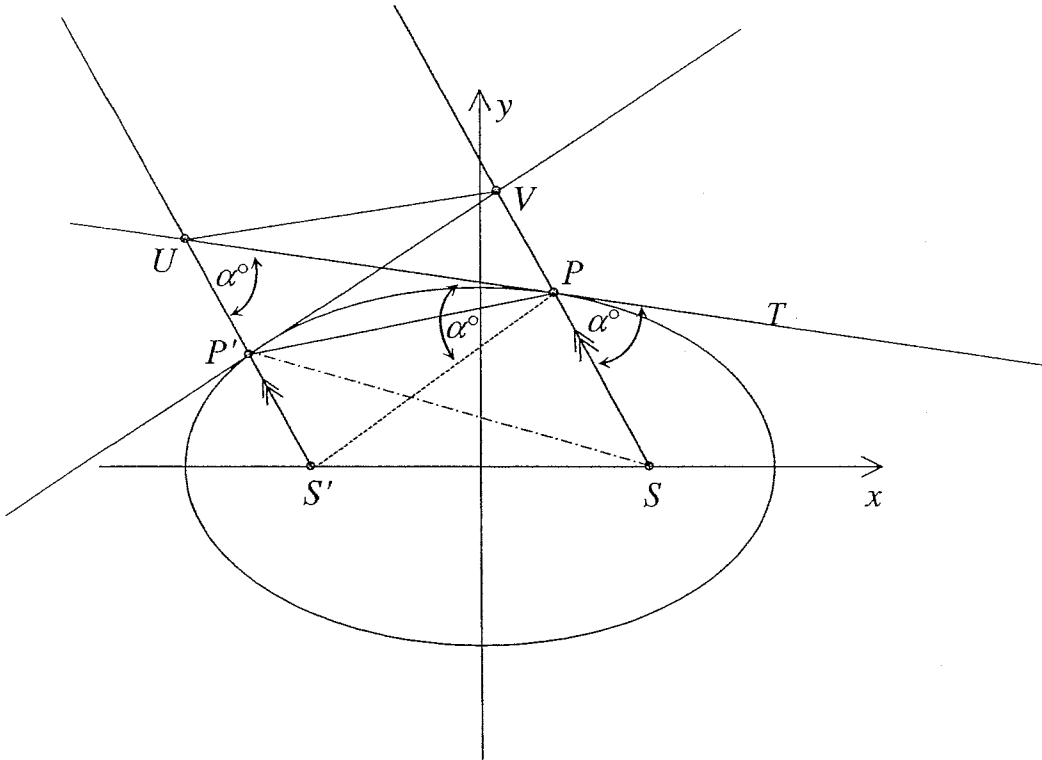
$$\therefore \frac{PM}{PM'} = \frac{PT}{PT'} \Rightarrow \frac{PM}{PT} = \frac{PM'}{PT'}$$

(iv) Using (ii) $\cos \alpha = \frac{SP}{PT} = \frac{ePM}{PT} = e \frac{PM}{PT}$ and similarly $\cos \beta = e \frac{PM'}{PT'}$

$$\therefore \cos \alpha = \cos \beta$$

$$\therefore \alpha = \beta \quad [\because 0 \leq \alpha, \beta \leq 90^\circ]$$

(c) (i)



(ii) From (b) $\angle UPS' = \angle SPT = \alpha$
 $\angle S'UP = \alpha$ (Corresponding angles are equal on parallel lines, $SV \parallel S'U$)
 $\therefore \triangle UPS'$ is isosceles.

(iii) Similarly $\triangle SP'V$ is isosceles
 $VP = VS - SP$

$$= SP' - SP \quad [\triangle SP'V \text{ isosceles}]$$

$$= SP' - (2a - S'P) \quad [\text{From (a)}]$$

$$= S'P - (2a - SP')$$

$$= US' - (2a - SP') \quad [\triangle UPS' \text{ isosceles}]$$

$$= US' - S'P' \quad [\text{From (a) but with } S'P' + SP' = 2a]$$

$$= UP'$$

(iv) $VP = UP'$; $VP \parallel UP' \Rightarrow UVPP'$ is a parallelogram
 $\therefore UV \parallel PP'$ (opposite sides of a parallelogram are parallel)

Question 8

$$\begin{aligned}
 \text{(a)} \quad \text{(i)} \quad \tan \left[\underbrace{\tan^{-1}(n+1)}_{\alpha} - \underbrace{\tan^{-1}(n-1)}_{\beta} \right] &= \tan(\alpha - \beta) \\
 &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\
 &= \frac{(n+1) - (n-1)}{1 + (n+1)(n-1)} \\
 &= \frac{2}{1 + n^2 - 1} = \frac{2}{n^2} \\
 \therefore \tan^{-1}(n+1) - \tan^{-1}(n-1) &= \tan^{-1}\left(\frac{2}{n^2}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad x > 1 &\Rightarrow \frac{\pi}{4} < \tan^{-1} x < \frac{\pi}{2} \\
 \therefore x > 1, y > 1 &\Rightarrow \frac{\pi}{4} + \frac{\pi}{4} < \tan^{-1} x + \tan^{-1} y < \frac{\pi}{2} + \frac{\pi}{2} \\
 \therefore \frac{\pi}{2} < \tan^{-1} x + \tan^{-1} y < \pi &\text{ i.e. } \tan^{-1} x + \tan^{-1} y \text{ lies in the second quadrant.}
 \end{aligned}$$

$$\begin{aligned}
 \text{BUT } x > 1, y > 1 \quad \frac{x+y}{1-xy} < 0 &\text{ and so } -\frac{\pi}{2} < \tan^{-1}\left(\frac{x+y}{1-xy}\right) < 0 \\
 \therefore -\frac{\pi}{2} + \pi < \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) < 0 + \pi &\Rightarrow \frac{\pi}{2} < \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) < \pi \\
 \text{So } \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) &= \sum_{r=1}^n [\tan^{-1}(r+1) - \tan^{-1}(r-1)] \\
 &= \left[\underbrace{\tan^{-1}(2) - \tan^{-1}(0)}_{r=1} \right] + \left[\underbrace{\tan^{-1}(3) - \tan^{-1}(1)}_{r=2} \right] + \left[\underbrace{\tan^{-1}(4) - \tan^{-1}(2)}_{r=3} \right] + \dots \\
 &\quad + \dots + \left[\underbrace{\tan^{-1}(n-1) - \tan^{-1}(n-3)}_{r=n-2} \right] + \left[\underbrace{\tan^{-1}(n) - \tan^{-1}(n-2)}_{r=n-1} \right] \\
 &\quad + \left[\underbrace{\tan^{-1}(n+1) - \tan^{-1}(n-1)}_{r=n} \right] \\
 &= \tan^{-1}(n+1) + \tan^{-1}(n) - \tan^{-1}(1) \\
 &= \pi + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \frac{\pi}{4} \\
 &= \frac{3\pi}{4} + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right)
 \end{aligned}$$

Question 8 continued

$$\begin{aligned}
 \text{(iv)} \quad \sum_{r=1}^{\infty} \tan^{-1}\left(\frac{2}{r^2}\right) &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) \\
 &= \lim_{n \rightarrow \infty} \left[\frac{3\pi}{4} + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) \right] \\
 &= \frac{3\pi}{4} + \lim_{n \rightarrow \infty} \left[\tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) \right] \\
 &= \frac{3\pi}{4} \quad \left[\because \lim_{n \rightarrow \infty} \frac{2n+1}{1-n-n^2} = 0 \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \text{(i)} \quad \text{LHS} &= S_k(x) - S_k(x+1) \\
 &= \frac{k!}{x(x+1)(x+2)\dots(x+k)} - \frac{k!}{(x+1)(x+2)(x+3)\dots(x+k+1)} \\
 &= \frac{k!(x+k+1) - xk!}{x(x+1)(x+2)(x+3)\dots(x+k+1)} \\
 &= \frac{k![(x+k+1) - x]}{x(x+1)(x+2)(x+3)\dots(x+k+1)} \\
 &= \frac{k!(k+1)}{x(x+1)(x+2)(x+3)\dots(x+k+1)} \\
 &= \frac{(k+1)!}{x(x+1)(x+2)(x+3)\dots(x+k+1)} \\
 &= S_{k+1}(x)
 \end{aligned}$$

(ii) Test $n = 1$

$$\text{LHS} = T_1(x) = \frac{{}^1C_0}{x} - \frac{{}^1C_1}{x+1} = \frac{1}{x} - \frac{1}{x+1} = \frac{1}{x(x+1)}$$

$$\text{RHS} = \frac{1!}{x(x+1)} = \frac{1}{x(x+1)}$$

So true for $n = 1$

$$\text{Assume true for some integer } n = k \text{ i.e. } T_k(x) = \frac{k!}{x(x+1)(x+2)\dots(x+k)}$$

$$\text{Need to prove it is true for } n = k+1 \text{ i.e. } T_{k+1}(x) = \frac{(k+1)!}{x(x+1)(x+2)\dots(x+k+1)}$$

Question 8 continued

$$\text{LHS} = T_{k+1}(x)$$

$$\begin{aligned}
 &= \frac{{}^{k+1}C_0}{x} - \frac{{}^{k+1}C_1}{x+1} + \frac{{}^{k+1}C_2}{x+2} - \dots + (-1)^k \frac{{}^{k+1}C_k}{x+k} + (-1)^{k+1} \frac{{}^{k+1}C_{k+1}}{x+k+1} \\
 &= \frac{{}^kC_0}{x} - \frac{({}^kC_1 + {}^kC_0)}{x+1} + \frac{({}^kC_2 + {}^kC_1)}{x+2} - \dots + (-1)^k \frac{({}^kC_k + {}^kC_{k-1})}{x+k} + (-1)^{k+1} \frac{{}^kC_k}{(x+1)+k} \\
 &= \frac{{}^kC_0}{x} - \frac{{}^kC_1}{x+1} + \frac{{}^kC_2}{x+2} - \dots + (-1)^k \frac{{}^kC_k}{x+k} \\
 &\quad - \left[\frac{{}^kC_0}{x+1} - \frac{{}^kC_1}{x+2} + \dots + (-1)^{k-1} \frac{{}^kC_{k-1}}{x+k} + (-1)^k \frac{{}^kC_k}{x+1+k} \right] \\
 &= T_k(x) - T_k(x+1) \\
 &= S_k(x) - S_k(x+1) \quad [\text{By assumption}] \\
 &= S_{k+1}(x) \quad [\text{From (i)}]
 \end{aligned}$$

So the formula is true for $n = k + 1$ when the formula is true for $n = k$.

So by the principle of mathematical induction the formula is true for $n \geq 1, n \in \mathbb{Z}$

(iii) Sub $x = \frac{1}{2}$ into both sides of

$$\begin{aligned}
 &\frac{{}^nC_0}{x} - \frac{{}^nC_1}{x+1} + \frac{{}^nC_2}{x+2} - \dots + (-1)^n \frac{{}^nC_n}{x+n} = \frac{k!}{x(x+1)(x+2)\dots(x+k)} \\
 &\therefore \frac{{}^nC_0}{\frac{1}{2}} - \frac{{}^nC_1}{\frac{1}{2}+1} + \frac{{}^nC_2}{\frac{1}{2}+2} - \dots + (-1)^n \frac{{}^nC_n}{\frac{1}{2}+n} = \frac{n!}{\frac{1}{2}(\frac{1}{2}+1)(\frac{1}{2}+2)\dots(\frac{1}{2}+n)} \\
 &\therefore 2 \left(\frac{{}^nC_0}{1} - \frac{{}^nC_1}{3} + \frac{{}^nC_2}{5} - \dots + (-1)^n \frac{{}^nC_n}{2n+1} \right) = \frac{n!}{\underbrace{\frac{1}{2}(\frac{3}{2})(\frac{5}{2})\dots(\frac{2n+1}{2})}_{n+1 \text{ terms}}} \\
 &\therefore 2 \left(\frac{{}^nC_0}{1} - \frac{{}^nC_1}{3} + \frac{{}^nC_2}{5} - \dots + (-1)^n \frac{{}^nC_n}{2n+1} \right) = \frac{2^{n+1}n!}{1 \times 3 \times 5 \times \dots \times (2n+1)} \\
 &\frac{{}^nC_0}{1} - \frac{{}^nC_1}{3} + \frac{{}^nC_2}{5} - \dots + (-1)^n \frac{{}^nC_n}{2n+1} = \frac{2^n n!}{1 \times 3 \times 5 \times \dots \times (2n+1)}
 \end{aligned}$$

End of solutions